The lady doth protest too much!

Inferences from (over-)informative arguments

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Abstract

Bayesian models of argument strength predict that a rational agent's belief in C should increase monotonically when more reasons are provided in favor of C. This study investigates a competing prediction: in natural language dialogues, arguments may feel weaker when *too many* reasons are provided. Evidence from extensive prior psycholinguistic experimentation shows that listeners expect speakers to produce informative utterances, and can make inferences to rationalize perceived redundancies. In this paper, we propose that pragmatic speakers prefer to provide "informative" reasons– reasons that would significantly increase a literal listener's credence from a disbelieving prior. Pragmatic listeners would then interpret excess reasons by inferring that the speaker expected him to be skeptical, which may in turn counteract the effectiveness of the argument. Using the Rational Speech Act framework, we develop a pragmatic model of argument strength that predicts the latter inference, and conduct a behavioral experiment to test its predictions against the first-order Bayesian model. We find that our model better predicts the data, and conclude that principles of resource rationality can help explain why we often prefer to trust each other, even in the absence of evidence.

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Chapter 1

Introduction

Language is built upon a foundation of trust, and there is a case to be made that this trust is rational. Our modern lives depend almost entirely on testimonial knowledge (Levy, 2021; Mercier and Sperber, 2017), and language use is evolutionarily stable only because cooperativity is generally advantageous (McCready, 2015). However, listeners obviously do not always automatically accept assertions on the basis of conversational norms (Farkas and Bruce, 2010; Sperber et al., 2010), and speakers do not expect their addressees to take implausible testimony for granted (Oey et al., 2023). Even when the maxim of Quality (Grice, 1989) is presumably in effect, interlocutors might have access to different bodies of evidence against which they evaluate facts, or else they might differ subjectively on matters of taste or opinion (Stephenson, 2007). This study concerns the use of *reasons* to negotiate controversial beliefs.

When some belief *C* is difficult to accept on the basis of testimony alone, a speaker may provide reasons to make it available by inference instead (Mercier, 2020; van Eemeren et al., 2014):

(1) You should take the bus to the office *because the subway is delayed by 45 minutes.*

Taking the bus to the office is ordinarily undesirable. However, once you know that the subway is delayed, you can easily infer that the bus is your best option. More generally, a reason *R* is said to "count in favor" of *C* when knowledge of *R* raises the probability of *C* in the context of an agent's other beliefs. Conclusions justified by many favorable reasons should then be more credible– or at least, no less credible– than conclusions justified by few or none. This analysis is intuitive, and it is predicted by standard Bayesian models of argument strength (Hahn and Oaksford, 2007; Oaksford and Hahn,

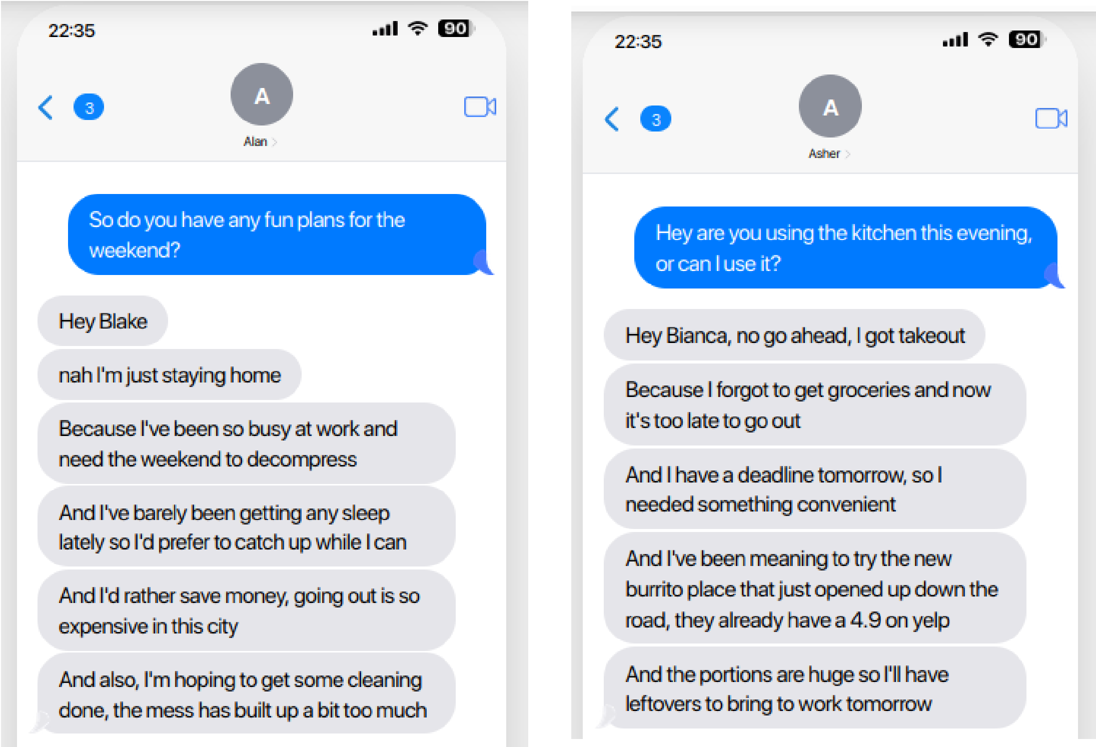


Figure 1.1: Too many reasons?

2004; Godden and Zenker, 2018). The present study, however, will investigate the competing intuition that superfluous reasons are somehow suspicious (e.g. Figure 1.1). Put famously by Queen Gertrude in Act III, Scene II of *Hamlet*: "The lady doth protest too much, methinks."

We aim to explain the latter intuition as an interaction between rational belief revision and rational language comprehension. A long tradition of psycholinguistic experimentation has demonstrated that speakers make "rational" usage of limited cognitive resources by constructing their utterances efficiently: we often reserve our longest signals for the least predictable content (Aylett and Turk, 2004; Levy and Jaeger, 2007). Comprehenders, in turn, have been shown to interpret utterances with an expectation for *informativity*, and often assume that speakers will not waste resources on content that is easy to predict (Rohde and Rubio-Fernandez, 2022; Sedivy, 2003). More recent work, following Grice (1989), investigates the sorts of inferences that listeners make to rationalize seemingly "over-informative" utterances. For example, apparently trivial script continuations can induce atypicality inferences (Kravtchenko and Demberg, 2022a,b):

(2) Alice went to the grocery store, and ***she paid the cashier!***

Assertion of (2) can lead the listener to believe that the continuation (Alice paying for her groceries) is actually an unusual or low-probability event.

Prior work on efficiency and informativity in pragmatics has typically been concerned with interlocutors' ability to coordinate on the *meaning* of utterances. The most informative utterances are just the ones which make the listener much more likely to infer which state of the world the speaker believes to be true (Degen, 2023). These models usually do not consider whether the listener will actually update his private information to align with the speaker's; in an ideally cooperative scenario, it may be taken for granted that he will. However, when speakers aim to influence listeners' beliefs by means of reason rather than trust, the most "informative" utterances are (as we will show) the ones that provide the strongest evidence for the conclusion– thus making it easier to actually believe or agree with.

With this definition of informativity in hand, we develop a pragmatic model for the production and interpretation of reasons, taking reasons for action as our case study. In our model, the speaker provides reasons to maximize the probability of the listener judging an action favorably, while also minimizing production cost. The model predicts that efficient speakers avoid giving reasons when the listener is likely to agree with her[1] in any case. Therefore, excess reasons can signal to a pragmatic listener that the speaker anticipated a disagreement. The listener might then search for a reason to doubt what he otherwise would not have thought to question. As such, reasons that *evidentially* count in favor of a conclusion can *pragmatically* count against it. We formalize this inference using the Rational Speech Act (RSA; Frank and Goodman, 2012) framework.

The experiment reported in the subsequent chapters investigates (1) whether listeners rationalize superfluous reasons by inferring that the speaker anticipated a disagreement, and (2) whether such an inference makes the reasons *less* convincing, not more. In a behavioral study, participants read a series of text-message conversations, in which a speaker justifies an action with zero or four reasons. In half the conversations, the justifications are provided out of the blue, and in the other half they are elicited discoursally to suppress the inference. For each conversation, participants rated whether the speaker thought the listener would respond favorably or unfavorably, and whether the listener would actually respond favorably. The results of the study generally support the prediction of Queen Gertrude: participants did infer that speakers had provided excess reasons because they expected a disagreement, and the inference also seemed to make the reasons less persuasive overall. Our findings suggest that the production cost

[1]For clarity, we refer to the speaker as "she" and the listener as "he" going forward.

of argumentation for cognitively bounded agents may help explain when and why trust can be rational.

Chapter 2

Background

2.1 Chapter roadmap

This preliminary chapter introduces two notions of rationality (hereafter "pragmatic" and "evidential") which make different predictions with respect to the evaluation of reasons. Section 2.2 introduces the pragmatic theory, which is concerned with efficient communication. In particular, we highlight recent work demonstrating that listeners are willing to adjust their model of the conversational context to make sense of utterances which might otherwise seem "over-informative" (Kravtchenko and Demberg, 2022a,b). Section 2.3 shows that these models do not capture the effect of reasons on an information state.

Section 2.4 introduces the standard Bayesian approach to argument strength (Hahn and Oaksford, 2007; Oaksford and Hahn, 2004), which cannot predict any inference in response to excess reasons. However, a pragmatic model in the style of Kravtchenko and Demberg (2022b), if adapted to handle information gained by reason, would be able to explain a "backfire" effect, if such an effect does arise.

2.2 A theory of rational language use

2.2.1 Rational speakers are informative

Theories of rational language use (Grice, 1989; Frank and Goodman, 2012; Levy and Jaeger, 2007) propose that speakers prefer to produce *informative* utterances. While "information" in pragmatics is conventionally a measure of logical strength, in the computational and psycholinguistic traditions, it is usually treated as a measure of surprise. On the latter accounts, all linguistic units from phonemes to full utterances are informative when surprising, and redundant when predictable (Levy and Jaeger, 2007). Following Shannon (1948), the information content ("surprisal") of a linguistic unit is quantified by its negative log-probability conditioned on the context:

$$I(u_i) = -\log P(u_i|u_0...u_{i-1}) \tag{2.1}$$

Shannon's model describes the optimal rate of information ("bit-rate") through a noisy channel with limited bandwidth. In order to maximize the amount of information that can be transmitted through the channel, a constant bit-rate that approaches, but does not exceed, the channel's capacity must be maintained. The most efficient coding scheme for a set of meanings M assigns $-\log P(m)$ bits to each $m \in M$. Or, less formally, longer signals are used for unpredictable content, and shorter signals for predictable content, in order to maintain the optimal bit-rate.

Natural languages are not optimal coding schemes, of course. However, it has long been of interest to linguists and psychologists whether speakers make "rational" usage of limited cognitive resources by constructing utterances efficiently (going back at least to Zipf, 1935). Over the past 20 years, numerous instantiations of the noisy-channel model have successfully captured patterns of signal reduction in low-surprisal contexts across various levels of linguistic structure. These include articulatory and durational phonetic reductions (Aylett and Turk, 2004; Bell et al., 2003; Pluymaekers et al., 2005), contractions (Piantadosi et al., 2011), lexical reductions (e.g. chimp/chimpanzee; Mahowald et al., 2013), complementizer omission (Jaeger, 2010), fragments (Bergen and Goodman, 2015; Lemke et al., 2021), and omission of discourse connectives (Torabi Asr and Demberg, 2012). All of these phenomena are shown to occur more frequently when the reduced or omitted content is contextually predictable.

Despite these findings, the "rational" theories (particularly Grice's maxim of Quantity) have previously come under fire because of their apparent inability to account for the logical redundancies that abound in naturalistic communication (Engelhardt et al., 2006; Heller et al., 2012). For example, speakers are willing to provide logically stronger references than strictly required to identify an object:

(1) *Context: The display contains just one square.*
"Look at the **blue square!**"

However, more recent work suggests that speakers are not "over-informative" as a matter of fact; rather, they prefer to overspecify in *high*-surprisal contexts. Extra modification occurs more frequently in complex visual displays (Rubio-Fernández, 2016), and in contexts where misunderstandings are expected (e.g. learner-directed speech; Bergey et al., 2020; Tal et al., 2023). It also tends to occur with unpredictable attributes ("yellow strawberry" rather than "yellow banana"), and with uniquely identifying or less ambiguous attributes (e.g. color rather than size) that make the identity of the upcoming referent less surprising (Degen et al., 2020).

2.2.2 Rational listeners expect informativity

If rational speakers are economical in this way, then rational listeners should interpret utterances with an expectation for informativity. While studies have historically shown that comprehenders tend to anticipate more plausible utterances (which are more likely to satisfy the Quality maxim; Bicknell et al., 2010), numerous others show that comprehenders do also expect to hear utterances that are informative or newsworthy (in keeping with Quantity-2; Hao et al., 2024; Rohde et al., 2021; Rohde and Rubio-Fernandez, 2022; Reksnes et al., 2024). For example, upon hearing the word "yellow", comprehenders expect the continuation to be something that is *plausibly* yellow (a notebook), but not *predictably* yellow (a banana), unless in the presence of an atypical contrastive object (a green banana; Sedivy, 2003).

This observed expectation for informativity raises the question of whether listeners simply tolerate redundancies, or else derive pragmatic inferences to rationalize them instead. Although some experiments show that redundant references can be tolerated, more expensive discourse-level inefficiencies are increasingly shown to generate inferences (Kravtchenko and Demberg, 2022a,b; Rees et al., 2026; Ryzhova et al., 2023). In particular, assertion of seemingly predictable content causes listeners to revise their background beliefs, adjusting their model of the context to make the utterance situationally informative. For example, Kravtchenko and Demberg (2022a) demonstrate that mentions of script continuations, which are generally available as default inferences, can give rise to an "atypicality" inference:

(2) Alice went to the grocery store, and ***she paid the cashier!***

Since "paying the cashier" would usually be trivially inferrable from world knowledge, listeners rationalize this type of utterance by assuming that Alice doesn't *normally* pay. Kravtchenko and Demberg (2022b) formalize this inference using the Rational

Speech Act framework (Frank and Goodman, 2012). The model developed in Chapter 3 is also a member of this family, so in the interest of conciseness, only a basic sketch is presented here, with further details reserved until then.

2.2.3 Inferences from over-informativity in the RSA framework

The RSA framework models pragmatic interpretation as theory-of-mind recursion. A listener infers the meaning s from utterance u by modeling the behavior of a speaker, who selects the best utterance u to convey meaning s by modeling the behavior of a listener, etc. In Kravtchenko and Demberg's model, the speaker selects an utterance $u \in \{\text{null}, \text{"She paid the cashier!"}\}$ to distinguish between meanings, i.e. states of the world $s \in \{\textbf{paid}, \neg\textbf{paid}\}$[1].

The recursion terminates with a "literal" version of the listener L_0, who performs no pragmatic reasoning. L_0 simply eliminates all s that do not satisfy the truth conditions of u (i.e. $\delta_{s \in [\![u]\!]} = 0$), and then weighs the remaining options ($\delta_{s \in [\![u]\!]} = 1$) according to their prior probability. Kravtchenko and Demberg (2022b) introduce an additional parameter h, representing the habituality of the action. Since a habitual action like paying the cashier in a store has a high prior $P(s|h)$, a literal listener will probably infer that the action took place, even if the speaker declines to specify it.

$$P_{L_0}(s|u,h) \propto \delta_{s \in [\![u]\!]} \cdot P(s|h) \tag{2.2}$$

A pragmatic speaker S_1 softmax-optimizes[2] her utterance, aiming to reduce the listener's surprisal at s, as well as the production cost of u:

$$P_{S_1}(u|s,h) = \text{softmax}(\alpha \cdot \log P_{L_0}(s|u,h) - \text{cost}(u)) \tag{2.3}$$

If the action is already predictable to L_0, then S_1 prefers the cheaper null utterance, since the risk of unclarity is minimal. Therefore, when S_1 *does* choose to mention the action, the pragmatic listener infers that this action is probably *not* habitual, since an efficient speaker is unlikely to have mentioned it if she assumed L_0 would infer it anyways:

$$P_{L_1}(s,h|u) \propto P_{S_1}(u|s,h)P(s|h)P(h) \tag{2.4}$$

[1]We adapt their notation slightly for consistency, but preserve the logic of their model.

[2]The softmax temperature α represents the extent to which S_1 chooses u "rationally".

It should be noted, however, that the atypicality inferences generated by over-informative assertions do not affect the posterior probability of s. L_1 infers that "paying the cashier" is *true*, despite also inferring that it is usually *unlikely* (i.e. he infers lower values of h). This is because literal listener eliminates from consideration all possible worlds that contradict u, the dynamic effect attributed to a successful assertion (Stalnaker, 1978). In the remainder of this chapter, we will direct our attention to discourses where the listener is not expected to update his beliefs in this way. In these discourses, we predict, a pragmatic reduction of the listener's prior may affect his posterior beliefs, too.

In the upcoming section, we begin by showing that the models we just explored were designed to quantify information gained by trust. We suggest that they are not (yet) suitable for quantifying information gained by reason.

2.3 Meaning, reasoning, and information

Most of the work we have reviewed so far is situated within the broader Gricean program, where the purpose of conversation is taken to be the cooperative exchange of knowledge. Listeners can acquire information from speakers just by understanding the "meaning" of an utterance (and updating their beliefs without further ado). With the cooperative principle firmly in place, the informational utility of a speaker's utterance only depends on the probability of a listener selecting the right meaning from a set of relevant alternatives, and the cognitive cost of production (Eq. 2.3, see also Degen, 2023 for an overview). This characterization often persists even in models of noncooperative discourse, allowing listeners to recover the speaker's intended meaning first, and assess its credibility thereafter (Cummins, 2025).

This notion of informativity has been highly theoretically useful, but we will suggest here that it is incomplete. Not all "communicated" information is acquired solely on the basis of meaning-recognition. Listeners are more likely to accept information that is more coherent or compatible with the rest of their beliefs, even if the source of that information is not particularly authoritative (Mercier, 2020; Sperber et al., 2010; Thagard, 2005), so when the listener has reason to doubt the speaker's testimony, she may guide him to agree with her nonetheless by increasing the contextual plausibility of her utterance instead. The purpose of *argumentation*, therefore, is to reduce or eliminate the need for trust among interlocutors by instead making the controversial content inferrable from other propositions ("reasons") which the listener either already believes, or will accept more easily (Mercier and Sperber, 2017; Mercier, 2020; Sperber et al., 2010).

As noted by Sperber et al. (2010), even Grice (briefly) concedes a distinction between information gained by trust, and information gained by reason:

> ***Conclusion of argument: p, q, therefore r (from already stated premises):*** *While U[tterer] intends that A[ddressee] should think that r, he does not expect (and so intend) A to reach a belief that r on the basis of U 's intention that he should reach it. The premises, not trust in U , are supposed to do the work* (Grice, 1989).

To illustrate the difference, consider this example adapted from Sperber et al., itself adapted from Grice:

(3) Andy: Steve doesn't have a girlfriend these days.
Barbara: He definitely does.
Barbara': He definitely does. He's been paying a lot of visits to New York lately.

In this modified example, Andy has already expressed doubt towards the conclusion that Steve has a girlfriend. He seems unlikely to change his mind just because Barbara says otherwise. Barbara cites the New York visits as as a *reason* to believe that Steve has a girlfriend, because Andy either already knows about the visits, or has no obvious reason to doubt them. If the visits are good evidence, Andy may infer that Steve probably has a girlfriend after all. If not, he will remain skeptical (perhaps Steve travels for work). But Barbara expects Andy to update his beliefs in light of the inference from **visits** to **girlfriend**, not because of *her own* belief in **girlfriend**.

However, the two versions of Barbara's utterance are equally "informative" with respect to $s =$ **girlfriend**, if we take informativity to be the listener's surprisal after eliminating all s that contradict $[\![u]\!]$, as do RSA models (Eq. 2.2). In both cases, that surprisal is 0 (both utterances would eliminate all the ¬**girlfriend** worlds). But there clearly is another sense in which these two are informationally distinct: in the second example, **girlfriend** is more coherent with– and predictable from– the rest of Andy's beliefs. Barbara might still select the latter utterance despite its higher production cost (for no extra benefit in terms of truth-conditional informativity) because it reduces the surprisal of **girlfriend** even *absent* the Stalnakerian update.

In the final section of this introductory chapter, we will introduce the model typically used to quantify the inferential/evidential effect of an utterance on an information state, and show that it cannot predict any "protest-too-much" inference. We then claim that the backfire effect can still be explained, however, if this "evidential" sense of informativity is also subject to the efficiency considerations we introduced in section 2.2.

2.4 Previous approaches to argument strength

Argumentation is a somewhat marginal case in the study of (cooperative) pragmatics, but it is of course a central concern for philosophers and psychologists. Although greater emphasis has traditionally been placed on deductive arguments, which are valid only if the conclusion is a logical consequence of the reasons, naturalistic arguments often fail to obtain this qualification (Mercier and Sperber, 2017; van Eemeren et al., 2014). It is intuitively the case that some informal arguments are far more psychologically compelling than others, though, so a gradient model of argument strength is required to capture this variability.

The strength of a reason is thus measured by its effectiveness in bringing about a desired belief. This approach lends itself to Bayesian models of rational belief update (Hahn and Oaksford, 2007; Godden and Zenker, 2018; Oaksford and Hahn, 2004). A set of reasons "counts in favor" of a conclusion if the posterior probability of the conclusion conditioned on the reasons is greater than the (pre-utterance) prior probability of the conclusion. The strength of a reason is defined as the extent to which it raises the posterior, as computed through application of Bayes' theorem:

$$\begin{aligned} P(C|R) &= \frac{P(R|C)P(C)}{P(R|C)P(C)+P(R|\neg C)P(\neg C)} \\ &\propto P(R|C)P(C) \end{aligned} \tag{2.5}$$

The Bayesian model has been put to use by linguists studying argumentation, particularly in the tradition of *Argumentation within Language* (Anscombre and Ducrot, 1983). Unlike mainstream semantics/pragmatics, this model actually takes the evidential potential of an utterance to be a primary semantic contribution (either in addition to, or instead of, the Stalnakerian context-change potential; see Iten, 2000; Winterstein, 2018 for overviews). The utility of an utterance, from the speaker's perspective, is given by its evidential support of a conclusion or "argumentative goal" upon conditionalization (Merin, 1999), not just by how precisely it specifies a state of the world (Eq. 2.3).

A tacit assumption of the Bayesian model is that, to a purely rational agent, the most convincing argument is the one that provides the best evidence for the conclusion. In fact, listeners may assume any potential "better" reasons that the speaker has declined to mention are false or unknown (analogous to a standard scalar implicature; Barnett et al., 2022; Cummins and Franke, 2021). While these models do predict “diminishing returns” on additional positive reasons when the prior is already high, they do not predict that excess reasons might actually count against the conclusion, since the Bayesian posterior increases monotonically with additional favorable evidence. Divergent behavior can be attributed to a failure of rationality (Hahn and Oaksford, 2007).

It is unclear, however, whether listeners really are always better off epistemically when the most conclusive evidence for a claim is provided. As we have seen, assertions are frequently accepted with no explicit evidence whatsoever. There is considerable debate among epistemologists as to whether the tendency to trust is justified (Lackey, 2008; Levy, 2021), if it exists at all (Brashier and Marsh, 2020; Mercier, 2020). Although we cannot not weigh in on that discussion here, we will make a related claim: language would surely be a highly *inefficient* means of accumulating knowledge if all information acquired testimonially had to be fully, or even mostly, inferrable from our other beliefs (even if language might also be more *reliable* in that case).

The model that we will propose in the next chapter predicts that arguments are produced and evaluated not just as abstract formal objects, but as resource-rational speech acts. If interlocutors optimize their utterances to maximize the coordination of their information states within cognitive constraints, speakers should provide more evidence in the case of strong disagreement, and no evidence in the case of agreement. If listeners expect this sort of economical behavior from speakers, an effortful argument for a seemingly agreeable conclusion should generate an inference that there might, in this case, be grounds disagreement after all, analogous to the inference described by Kravtchenko and Demberg (2022a). And further, that inference might undercut the actual evidential strength of the reasons.

Chapter 3

Models

3.1 Chapter overview

The objective of our study is to investigate whether justifying a conclusion with many reasons enhances or undermines an argument, and to determine whether this effect depends on the (over)informativity of the reasons. In this chapter, we compare the qualitative and quantitative predictions of two probabilistic models, which correspond to our "evidential" and "pragmatic" hypotheses. The experiment reported in the remaining chapters is designed to distinguish between these two hypotheses:

- The **evidential** model is a simple, first-order Bayesian belief update. This model predicts that agents' credence should increase monotonically as additional favorable evidence for the conclusion becomes available.
- The **pragmatic** model embeds basic evidential reasoning within a higher-order social inference, adapting the Rational Speech Act framework. First, we show how the usual RSA setup can be adjusted to model evidential informativity, rather than truth-conditional informativity. Then, we show that agents who apply both evidential and social reasoning when interpreting argumentative utterances may not always increase their credence upon hearing additional favorable reasons. When the speaker is willing to expend substantial production cost to justify a conclusion, her utterance signals to the listener that she anticipated a strong disagreement. That signal may be enough to counteract the evidential contribution of the reasons.

Before we proceed, two points of clarification are in order. Firstly, as discussed by many theorists in the tradition of argumentation studies, the notion of an argumentative "goal" or "conclusion" can be quite broad. Reasons are readily used to justify any type of psychological effect on which interlocutors might attempt to coordinate: factual beliefs, decisions and actions, preferences, emotional attitudes, etc. (Mercier and Sperber, 2017). Our study focuses on reasons that justify actions, using stimuli in the form: "*I did [action] because [reasons].* " Agents use reasons to judge actions' normative (i.e. deliberative/deontic) status (Alvarez and Way, 2024; Kearns and Star, 2009), so we will consider an argument to be effective if it causes an agent to believe that the action mentioned by the speaker is *better* or *more favorable*. Nothing about the basic logic of our model is exclusive to judging actions, so the framework is easily adapted to suit reasons for other psychological effects. However, experimentally verifying the model's predictions for other types of argumentative goals will have to wait for another time.

Secondly, we want to be sure that any "backfire" effect we might observe in our experiment is actually due to pragmatic reasoning about the speaker's choice to produce an expensive utterance. Therefore, the inference should disappear in discourses where it is sensible for the speaker to provide multiple reasons even if she thinks the listener is generally inclined to agree with her (or in other words, the listener's prior does not play a significant role in the speaker's choice to elaborate). We will expand further on the contextual variation in the description of our materials in Chapter 4, but briefly, in these "prompted" contexts, the listener is curious to hear more (e.g. "tell me more!") about an action that typically involves weighing multiple factors (e.g. moving to a new neighborhood).

3.2 Notation

Let $\mathcal{V} : A \times W \to [0,1]$ be a decision-theoretic *reward function*, which accepts an action $a \in A$ and a possible world $w \in W$, and returns a scalar *value* $v \in V$, representing how good or favorable action a would be if the true state of the world were w. We designate the unit interval $V = [0,1]$ as the scale of values an action may take at any given world, with 0 representing the least favorable actions, and 1 representing the most favorable actions. In standard decision theory, reward functions are unbounded (i.e. $V = \mathbb{R}$; Sumers et al., 2023; Sutton and Barto, 2018), but for ease of modeling, we simplify to the unit interval here.

Following Achimova et al. (2025), we model an agent's prior belief about the given action a as a probability density function π over the value scale $V = [0,1]$. When π places greater probability density on low values of v, the agent has an unfavorable prior opinion. He believes that the true state of the world is most probably one where a is unfavorable or dispreferred. Likewise, when π places greater probability density on high values of v, the agent has a favorable prior opinion.

For simplicity, we constrain the set of possible priors π to beta distributions, parameterized by mean μ and a fixed sample size (here we choose a moderate value of 10). The mean (expected value) μ of this distribution corresponds to the agent's judgment under uncertainty of the action's deliberative or deontic status (Lassiter, 2017), while the sample size represents how confidently the listener has made that judgment. We limit the speaker's utterance R to a binary choice on whether or not to provide reasons (in our experiment stimuli, the number of reasons was either 0 or 4). Further, since we are only interested in reasons that count in favor of a, we select a positive linear likelihood $p(R|v)$ for the "long" ($R = 4$) utterance, and a flat likelihood for the "null" ($R = 0$) utterance, such that the long utterance shifts the posterior to more favorable v, and the null utterance reproduces the prior. More conceptually, a positive linear likelihood $p(R|v)$ indicates that favorable reasons are more likely to be true in the high-action-value worlds (e.g. "the subway is delayed" is more likely to be true in worlds where the bus is a good option, vs worlds where the bus is a bad option). Limitations of these modeling choices are discussed in Chapter 6.

We use the notation π_μ to represent a beta prior with mean μ (and sample size 10). Example priors are given in Figure 3.1:

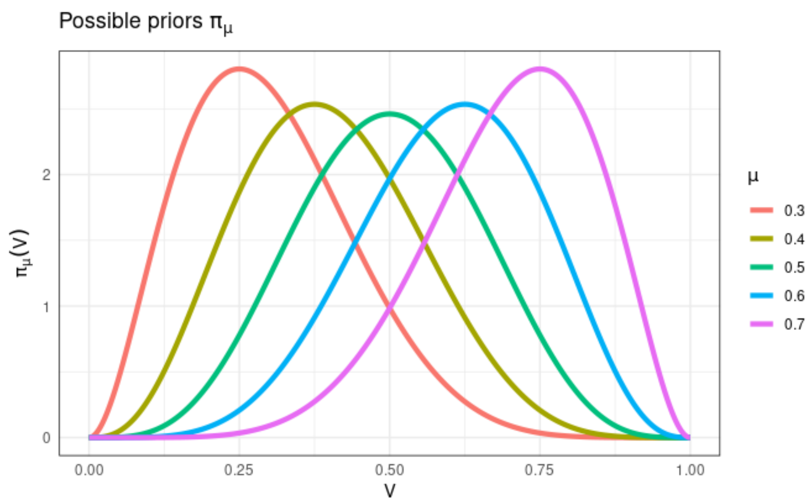


Figure 3.1: Some possible priors π_μ with sample size 10, ranging from unfavorable (red, $\mu = 0.3$) to favorable (purple, $\mu = 0.7$).

3.3 Research questions

We distinguish between the "evidential" and "pragmatic" models based on their predictions with respect to the following two questions. We will also investigate whether the effects are moderated by context ("prompted" or "unprompted").

1. Will the listener simply tolerate superfluous reasons? Or will he rationalize such utterances by inferring that the speaker expected him to disagree with her? In other words, does the listener infer from excess reasons that the speaker expected him to have an unfavorable prior π_μ?

2. Assume that the listener does indeed infer from the excess reasons that the speaker expected him to have an unfavorable prior (as per RQ1). Does that unfavorable prior negatively impact his own posterior belief $p(v|R)$?

3.4 Evidential model: more is more

We will begin by introducing the "evidential" model, which is standard in argumentation studies. This model predicts that the listener will update his belief in v (favorability of the action) in response to utterance R (0-reason or 4-reason explanation) just by conditioning on the new evidence. We will use the notation L_0^μ for an evidential listener whose prior is π_μ. An evidential listener's posterior belief $p_{L_0^\mu}(v|R)$ is calculated through straightforward application of Bayes' theorem[1]:

$$p_{L_0^\mu}(v|R) \propto p(R|v)\pi_\mu(v) \tag{3.1}$$

The evidential listener L_0^μ does not make any pragmatic inferences about why the speaker has chosen to provide or forego the reasons, so the "long" utterance does not signal to him that the speaker anticipated a disagreement (RQ1). His posterior $p_{L_0^\mu}(v|R)$ can only increase when conditioning on additional favorable reasons (RQ2), although the magnitude of the increase will be small when his prior is already favorable (see Figure 3.2). Further, since L_0^μ only thinks about the evidence itself, not the speaker's motive for providing it, his judgments should be the same in the "prompted" and "umprompted" contexts.

[1]Going forward, we use uppercase for discrete variables and their choice probabilities, and we use lowercase for continuous variables and their probability density functions.

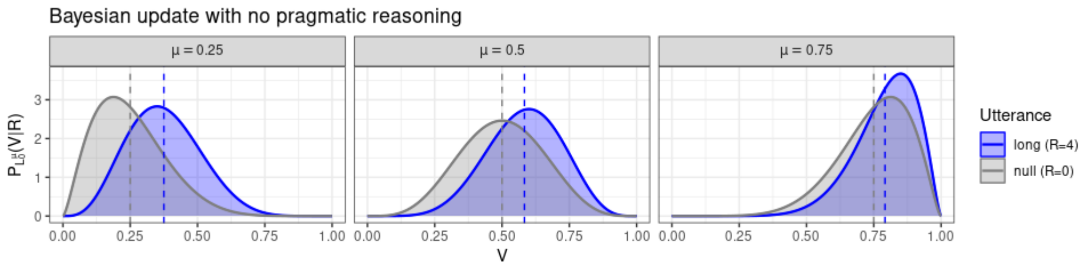


Figure 3.2: Posterior density $p_{L_0^\mu}$ for an evidential/literal listener. The long utterance ($R = 4$) always improves upon the null ($R = 0$), but the effect is smaller for more favorable priors.

3.5 Pragmatic model: less is more

Finally, we will introduce our new "pragmatic" model, which balances the evidential strength of the utterance R against an RSA-style social inference about the speaker's choice to utter it. As we step through the three RSA agents, we will highlight the key differences between our model and the traditional interpretation-only models.

3.5.1 Literal listener

As we previously described, RSA models terminate their recursive social reasoning with a base-case "literal listener", who updates his beliefs based solely on his prior and the literal semantics of the utterance. In the standard recipe, he infers the most likely *meaning* of utterance u by eliminating from consideration all possible meanings that do not satisfy its truth-conditions, and then renormalizing the prior over the remaining options (in a probabilistic adaptation of Stalnaker, 1978):

$$P_{L_0}(s|u) \propto \delta_{s \in [\![u]\!]} \cdot P(s) \tag{3.2}$$

When evaluating reasons, however, the listener does not just update his belief by inferring what the speaker's utterance means (for simplicity, we will also take it for granted that the meaning of R is unambiguous). Instead, he updates his belief in v based on the evidential strength of the given reasons R, just like our evidential listener (repeated from Eq. 3.1):

$$p_{L_0^\mu}(v|R) \propto p(R|v)\pi_\mu(v) \tag{3.3}$$

To illustrate the difference, let us revisit the bus/subway example. Our evidential listener hears an utterance of *The subway is delayed*, and determines that this is most likely to be true in worlds where taking the bus is favorable. Therefore, he updates his beliefs by increasing the probability of higher values of v, and decreasing the probability of lower values of v. Our model represents learning by reason, or inference from evidence. In the standard model, however, L_0 would hear an utterance like *Taking the bus is great!*, eliminate all values of v which fail to verify that utterance, and finally, redistribute his prior amongst the remaining values of v (presumably the upper range, in this case). Hence, the typical RSA L_0 can learn (approximately) the same information as ours, but by means of trust in the speaker's testimony.

3.5.2 Pragmatic speaker

Our pragmatic speaker S_1 faces the same decision problem as the pragmatic speaker described by Kravtchenko and Demberg (2022b). She selects an utterance to maximize the information gained by the literal listener, at minimal cognitive cost. The utility of an utterance R to a speaker whose prior is π_μ is given as:

$$\mathbb{U}_{S_1}(R;\mu) = \inf(R;\mu) - \text{cost}(R) \tag{3.4}$$

Bear in mind, though, that Kravtchenko and Demberg are interested in truth-conditional informativity, so their pragmatic speaker decides how precisely she needs to *specify* the true state of the world (e.g. that Alice paid the cashier) in order for L_0 to understand it. By contrast, our speaker decides how much evidence is required to support the conclusion, in order for L_0 to agree with it. More formally, $\inf(R;\mu)$ in our model is the extent to which conditioning on R will shift π_μ towards more favorable values of v. The Shannon information gained by conditionalization is standardly modeled by the KL-divergence between the posterior and the prior (Kullback and Leibler, 1951):

$$\inf(R;\mu) = D_{KL}(p_{L_0^\mu}(v|R)||\pi_\mu(v)) \tag{3.5}$$

Since $R = 4$ has a higher production cost than $R = 0$, the speaker would prefer to provide the long utterance only when she assumes she is talking to a literal listener with a skeptical or unfavorable prior. When L_0's prior is unfavorable, $R = 4$ provides a significant information gain from prior to posterior, which makes the more expensive utterance worthwhile (see again Figure 3.2). However, much like Kravtchenko and Demberg's speaker, if our speaker thinks the literal listener *already* has a favorable prior, the informativity of the long utterance is negligible, so the more economical null has greater utility.

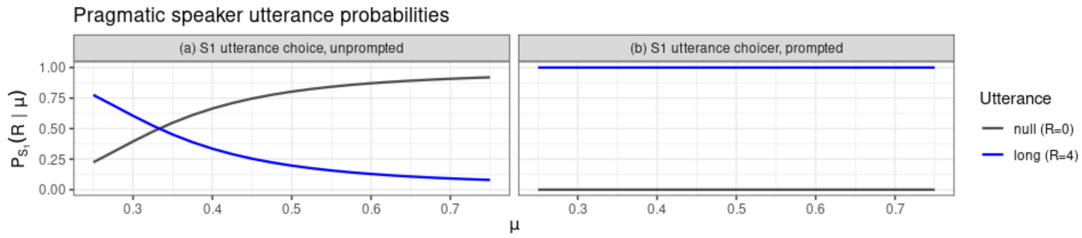


Figure 3.3: Utterance choice probabilities $P_{S_1}(R|\mu)$ for a pragmatic speaker. In unprompted contexts, the speaker prefers to give many reasons when she expects the listener will have an unfavorable prior, and no reasons otherwise. In prompted contexts, she strongly prefers to give reasons, regardless of of the listener's prior.

This formulation corresponds to unprompted reasons, which have no contextually obvious utility besides inducing a more favorable v. However, if the listener has prompted the speaker to elaborate, it makes sense for the speaker to provide the reasons no matter what she thinks his prior is, so $R = 4$ should always have higher utility. In either case, the speaker selects an utterance by softmax-optimizing her utility:

$$P_{S_1}(R|\mu) = \text{softmax}(\alpha \cdot \mathbb{U}_{S_1}(R;\mu)) \tag{3.6}$$

The softmax temperature α represents the extent to which the speaker selects her utterance "rationally" (if $\alpha = 0$, the speaker chooses randomly, and as $\alpha \to \infty$, the speaker chooses the highest utility utterance every time). The pragmatic speaker's utterance choice probabilities for $\alpha = 10$, $\text{cost}(\textit{null}) = 0$, and $\text{cost}(\textit{long}) = 0.3$ are plotted in Figure 3.3.

3.5.3 Pragmatic listener

In standard RSA models, a pragmatic listener L_1 infers which state of the world the speaker believes to be true, under the assumption that she has selected her utterance rationally (i.e. by optimizing its utility as per Eq. 3.6). Our L_1 does this too: he infers whether the speaker believed L_0 would agree or disagree with her.

$$p_{L_1}(\mu|R) \propto P_{S_1}(R|\mu) \cdot p_{L_1}(\mu) \tag{3.7}$$

L_1 applies Bayes' theorem to determine what sort of prior the speaker expected L_0 to have, based on her utterance R, and his prior-over-priors ("hyperprior"). The pragmatic listener's hyperprior over possible priors is notated as $p_{L_1}(\mu)$. The likelihood $P_{S_1}(R|\mu)$ of a pragmatic speaker producing utterance R, given that she expected L_0 to have the prior π_μ, is denoted by Eq. 3.6 (which we have already seen, but reproduced here):

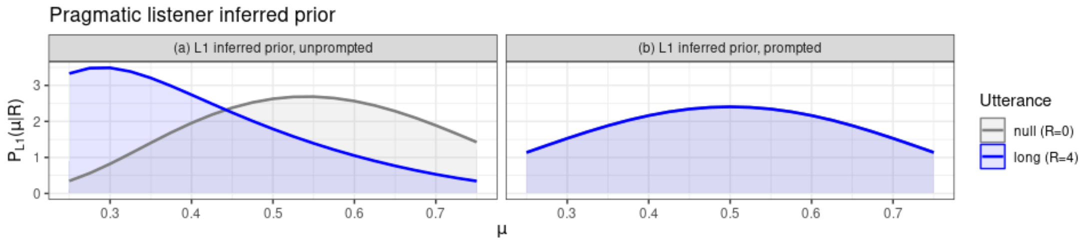


Figure 3.4: The pragmatic listener infers which prior π_μ the speaker expected him to have. In unprompted contexts, L_1 infers lower μ from many reasons, and higher μ from no reasons. In prompted contexts, the utterance choice does not impact his inference of μ.

$$P_{S_1}(R|\mu) = \text{softmax}(\alpha \cdot \mathbb{U}_{S_1}(R;\mu)) \tag{3.8}$$

This inference determines our model's predictions for RQ1. In the unprompted case, we saw that S_1 prefers the long utterance when she anticipates unfavorable priors (low μ), and the null utterance when she anticipates favorable priors (high μ). Therefore, our L_1 infers lower values of μ after hearing the long utterance, and higher values of μ after hearing the null utterance. Less formally, when the speaker forgoes reasons, the listener infers that she expected him to agree. When she provides reasons, the listener infers that she expected him to *dis*agree.

In the prompted case, however, S_1 always prefers the long utterance independently of μ. Her utterance therefore cannot help L_1 infer μ, and his hyperprior $p(\mu)$ alone determines his prediction. These predictions are plotted in Figure 3.4.

In the basic RSA setup, the pragmatic listener's goal is to recover the meaning of an utterance, so his job is done once he infers which state of the world the speaker believes to be true. Our listener still has work to do, because he updates his belief in v based on both the speaker's intentions (like standard interpretive RSA listeners) and the evidential strength of the reasons themselves (like our evidential listener).

Therefore, just like the evidential/literal listener L_0^μ, the pragmatic listener L_1 also updates his distribution over v by conditioning on R. However, instead of computing the posterior by conditioning his *original* prior on R, he takes a weighted average of the posteriors obtained by conditioning each possible prior π_μ on R. Each prior is weighted according to $P_{L_1}(\mu|R)$ which, as given by Eq. 3.7, takes into consideration both the listener's independent beliefs (the hyperprior) and the speaker's utterance choice probability $P_{S_1}(R|\mu)$.

$$p_{L_1}(v \mid R) \propto \int_0^1 \underbrace{p(R \mid v)\pi_\mu(v)}_{\textbf{Conditionalization from prior } \pi_\mu} \times \underbrace{p_{L_1}(\mu \mid R)}_{\textbf{Probability of that prior}} d\mu \tag{3.9}$$

Here $p(R \mid v)\pi_\mu(v)$ represents the posterior probability $p_{L_1}(v|R)$ if the "true prior" is π_μ, and $p_{L_1}(\mu \mid R)$ is the probability of π_μ being the "true prior".

The posterior probability $p_{L_1}(v|R)$ corresponds to our model's predictions for RQ2. In the unprompted case, since L_1 tends to place greater probability on less favorable priors upon hearing the long explanation, the evidential strength of the reasons themselves may not be sufficient to recover from the reduction of the prior. L_1's posterior may then be *lower* after hearing the long explanation, as compared to the null. In the prompted case, however, since $p_{L_1}(\mu \mid R)$ is the same for $R = 0$ and $R = 4$ (i.e. just L_1's hyperprior), the evidential strength of the long explanation always yields a greater posterior than the null. These predictions are plotted in Figure 3.5.

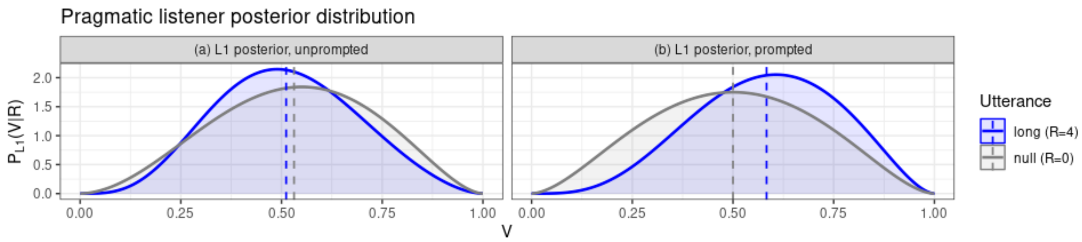


Figure 3.5: Posterior distribution $P_{L_1}(v|R)$ for a pragmatic listener. In only the unprompted contexts, the reasons may "backfire" and reduce the posterior.

Summing up, we have sketched out two different ways in which a listener might update his beliefs in response to reasons, and formalized each of these predictions with a probabilistic model. The "evidential" model predicts that the listener will only consider the evidential strength of the reasons themselves. The "pragmatic" model predicts that he will consider the evidence in the context of a rational speaker's choice to provide it. To this end, we also show how to modify the RSA framework to account for information gained by reason ("showing"), rather than by trust ("meaning"). Ultimately, the evidential model predicts that favorable reasons can only increase the listener's posterior, whereas the pragmatic model predicts that favorable reasons might decrease his posterior in some cases.

Chapter 4

Methods

We conducted a behavioral study to compare the predictions of the evidential and pragmatic models on (1) whether mentioning reasons for actions can generate an inference about the speaker's model of the listener's prior judgment of that action, and (2) whether such an inference can reduce the listener's posterior judgment of the action.

4.1 Participants

60 self-identified native English speakers residing in the United States or United Kingdom were recruited through Prolific. An additional 6 participants were recruited for a shorter pilot study; their data is not included in the analysis. All participants in the main study passed the attention checks, and were therefore included in the analysis. The survey took a median time of 17 minutes, 52 seconds to complete, and participants were compensated £5.00.

4.2 Study Design

The experiment used a 2x2 Latin square design, crossing the number of REASONS given to justify an action (**null-0R**, **long-4R**) with the CONTEXT in which the reasons were given (**unprompted**, **prompted**). The number of reasons was manipulated within items, and the context was manipulated between items (with 8 different items in each type of context).

4.3 Materials

Prior work has shown that speaker salience increases listeners' expectation for informativity (Reksnes et al., 2024), so the stimuli were presented as text message conversations between two named characters (following Wallbridge et al., 2021). To avoid confusion, the "explainer" character's name always began with A, and the "addressee" character's name always began with B.

Since the RQs target listeners' inferences about speakers' production choices, the explanation always came from the "incoming"/grey bubble character (A), and the two rating scale questions concerned the opinion of the "outgoing"/blue bubble character (B), thus allowing the participant to assume the role of the listener/addressee.

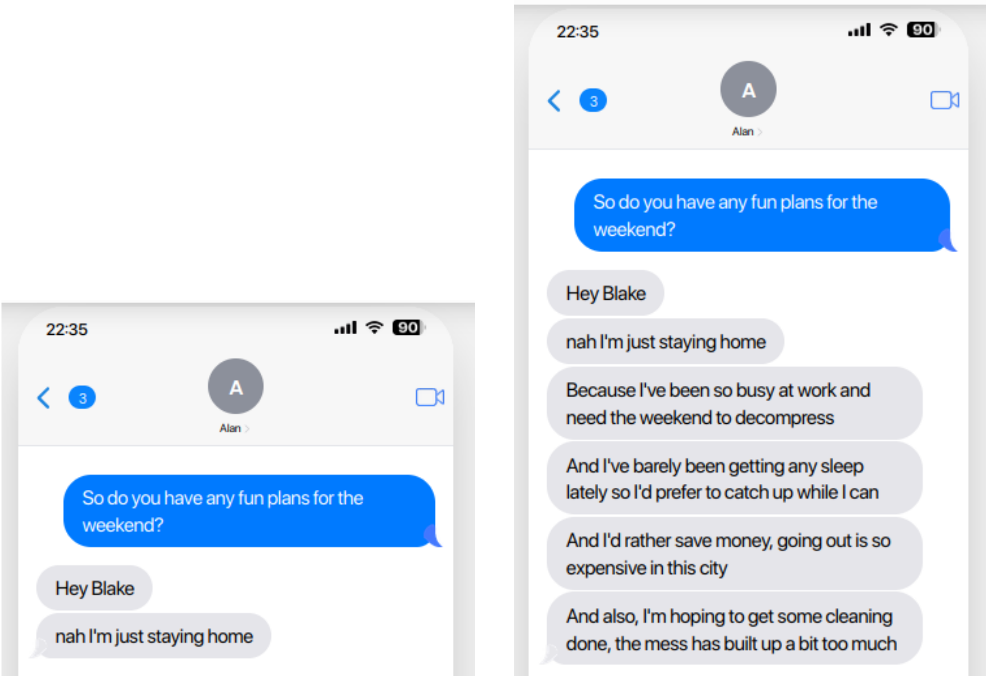
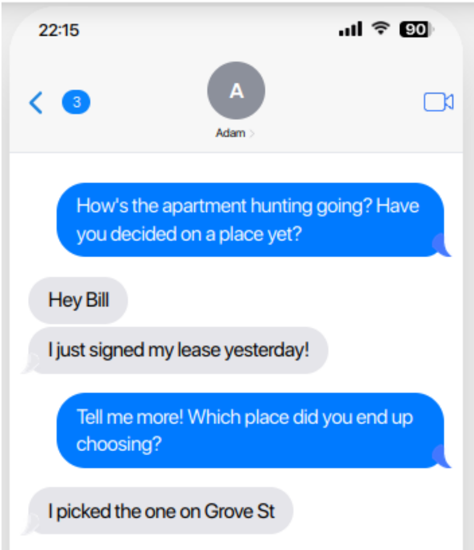


Figure 4.1: Sample **unprompted** context stimuli in **null-0R** and **long-4R** conditions.

The conditions were constructed as follows:

- In the **null-0R** condition, character A mentions the action but gives no reasons.
- In the **long-4R** condition, character A gives four reasons. The individual reasons were not held constant across items, but were instead tailored to provide good justification for each individual action (to avoid confounding with the "weak evidence" effect e.g. Barnett et al., 2022; as well as to maintain the naturalism of the dialogues).



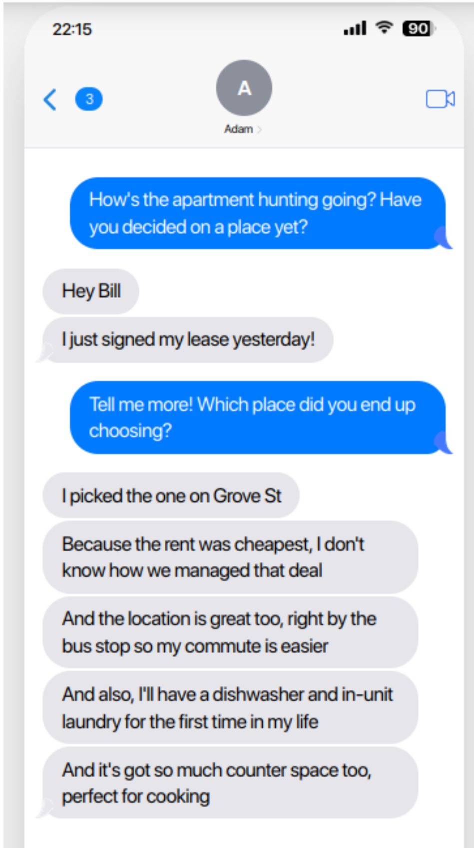


Figure 4.2: Sample **prompted** context stimuli in **null-0R** and **long-4R** conditions.

- For the **unprompted** items, character A describes an "ordinary" or everyday action (e.g. cleaning the kitchen or studying in the library), where participants were expected to have generally favorable priors ($\mu > 0.5$).
- For the **prompted** items, character A describes an action that would involve a deliberative process where multiple reasons would be considered (e.g. moving to a new city or making an expensive purchase), in response to a "which one"/"tell me more"-type prompt from character B (where giving many reasons is appropriate regardless of μ).

All stimuli were approximately the same length per condition, again with some variability for naturalism.

On each trial, participants responded to two questions on a seven-point Likert scale ("very unfavorable" to "very favorable"):

1. First, before sending the texts, did A expect B to view her decision unfavorably, or favorably? A expected B to feel. . .

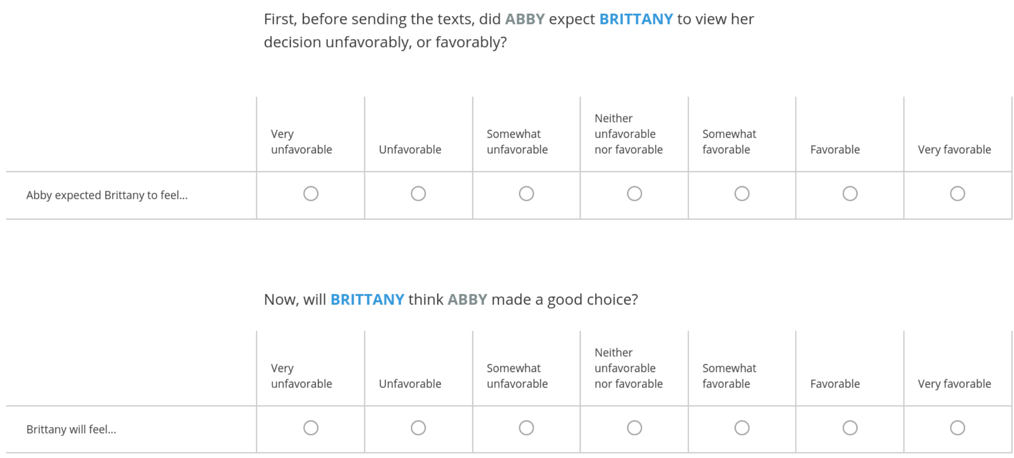


Figure 4.3: Sample trial.

2. Now, will B think A made a good choice? B will feel...

The first rating was intended to target RQ1: whether the listener would think the speaker gave reasons because she expected that he would otherwise disapprove. The second rating was intended to target RQ2: whether the listener's recognition of that expectation could counteract the effectiveness of the reasons in influencing his beliefs. In other words, the first question identifies whether the listener infers lower values of μ in the **long-4R** condition than in the **null-0R** condition, and whether that effect is moderated by the context. The second question identifies whether a reduction in the prior can also yield a reduction in the posterior $p(v|R)$ in the **long-4R** condition (compared to the **null-0R** condition), and again, whether that effect is moderated by the context.

For items in the **long-4R** condition, participants were also asked to provide a free-text response to the question "Why do you think A explained herself?", to see if participants might have an alternative rationalization for the over-informativity.

In addition to the target trials, participants saw 4 attention checks where character B responds to character A with explicit approval or disapproval, so the participant could trivially identify his opinion (in the target trials, the messages cut off before B's response, requiring the participants to make an inference to judge his opinion).

4.4 Procedure

The experiment was hosted online on Qualtrics. Before proceeding to the survey, participants were asked to read an information sheet and consent to participation (approved by the University of Edinburgh LEL Research Ethics Board) and to confirm their Prolific ID and native-speaker status. The participants were then instructed to read each set of messages closely and reason about the characters' motives for sending them.

Using the Qualtrics survey randomizer, each participant was assigned to one of two stimulus lists. Each stimulus list contained 4 target trials in each of the 4 conditions, and 4 attention checks, for a total of 20 trials. Each item appeared in only one condition per list, and participants saw the items in random order. Each trial was presented on a separate page, and participants were required to answer each of the questions before clicking to advance. At the end of the survey, participants completed a "debrief", where they were asked to provide feedback and guess what the experiment was about.

Chapter 5

Results

5.1 Model selection

Analysis for the Likert rating questions was conducted using a Bayesian cumulative logit model, as is best practice for ordinal data (Bürkner and Vuorre, 2019), using the R (R Core Team, 2021) package `brms` (Bürkner, 2017) with default priors. The fixed effects included REASONS (**null-0R** vs **long-4R**), CONTEXT (**unprompted** vs **prompted**), and their interaction, treating **null-0R** and **unprompted** as the baseline conditions. To account for participant and item-level variability, random effects for PARTICIPANT and ITEM were also included.

5.2 RQ1: Reduction of the prior

The first Likert rating question (*Before sending the texts, did A expect B to view her decision unfavorably, or favorably?*) targets RQ1: whether the listener would attempt to rationalize the speaker's choice to (over)explain herself, by inferring that she expected him to have an unfavorable prior. Responses are plotted in comparison to model predictions in Figure 5.1.

The model identified a credible main effect of REASONS (β = -0.77, 95% CI -1.09, -0.43), indicating that participants expected *less* favorable priors when *more* reasons were given in **unprompted** contexts. There was also a credible interaction of REASONS with CONTEXT (β = 0.83, 95% CI 0.35, 1.30), indicating that the reasons had no effect on the listener's inferred prior for **prompted** items. There was also no credible effect of CONTEXT for the **null-0R** items, indicating that **prompted** and **unprompted** items were rated about the same when no reasons were provided. Results of the models are

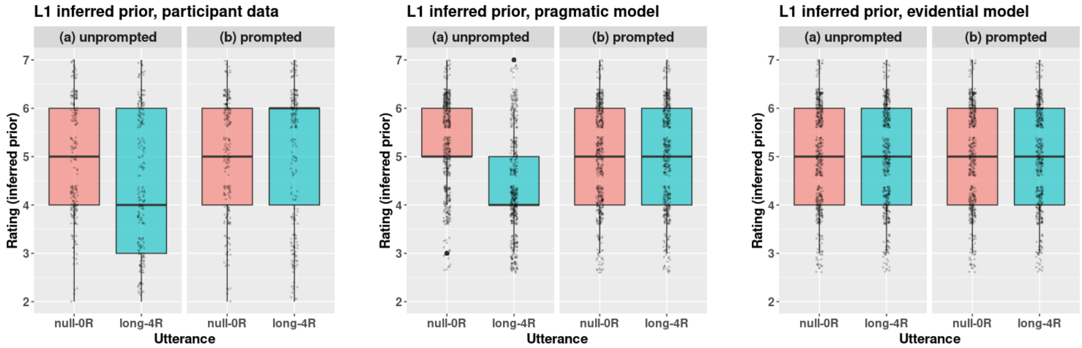


Figure 5.1: Participant responses to Q1 (left), compared to 500 simulated draws from the pragmatic (center) and evidential (right) models, using the brms intercepts as rating thresholds. The pragmatic model correctly predicts a negative effect of reasons on the inferred prior in unprompted contexts.

Predictor	Estimate	Std. Error	95% CI Lower	95% CI Upper
LONG4R	**-0.77**	**0.17**	**-1.09**	**-0.43**
PROMPT	0.19	0.66	-1.12	1.52
LONG4R:PROMPT	**0.83**	**0.24**	**0.35**	**1.30**

Table 5.1: Log-odds coefficients from the Bayesian ordinal model of Q1. A **credible** main effect of REASONS, moderated by a **credible** interaction with CONTEXT, is observed.

summarized in Table 5.1.

These results are in keeping with the pragmatic model. Participants rationalized the speaker's decision to provide **unprompted** reasons by inferring that the speaker expected the listener to have an unfavorable prior (i.e. he would need to be *convinced* to share her opinion). The evidential hypothesis would instead predict no effect for either context, since the listener does not make an inference about the speaker's choice to provide reasons in any case.

5.3 RQ2: Reduction of the posterior

The second Likert rating question (*Now, will B think A made a good choice?*) targets RQ2: whether higher-order inferences about *why the speaker felt an explanation was necessary* would mitigate that explanation's effectiveness in persuading the listener (i.e. reduce its argument strength). Responses are plotted in comparison to model predictions in Figure 5.2.

There was no credible main effect of REASONS for **unprompted** items, indicating that the participants rated those items about the same, whether or not the speaker provided reasons. By contrast, there was a credible interaction of REASONS with

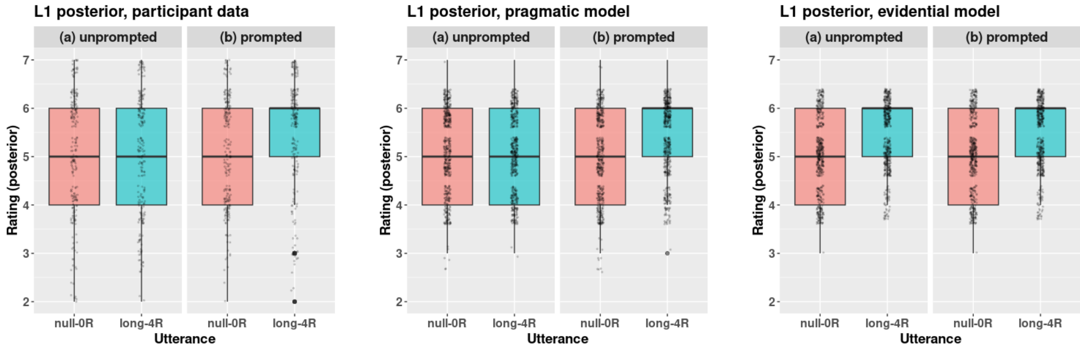


Figure 5.2: Participant responses to Q2 (left), compared to 500 simulated draws from the pragmatic (center) and evidential (right) models, using the brms intercepts as rating thresholds. In the pragmatic model and the participant data, the reasons positively influence the posterior only in the prompted context, due to the reduced prior in the unprompted context. In the evidential model, the reasons positively influence the posterior in both contexts, since there is no reduction of the prior in either context.

Predictor	Estimate	Std. Error	95% CI Lower	95% CI Upper
LONG4R	0.15	0.17	-0.20	0.50
PROMPT	0.25	0.76	-1.25	1.79
LONG4R:PROMPT	**0.75**	**0.24**	**0.28**	**1.22**

Table 5.2: Log-odds coefficients from the Bayesian ordinal model of Q2. A **credible** interaction of REASONS and CONTEXT is observed.

CONTEXT ($\beta = 0.75$, 95% CI 0.28, 1.22), meaning that participants rated the **prompted** items higher when the speaker provided reasons than when she did not. There was still no credible main effect of CONTEXT, indicating that **prompted** and **unprompted** items were rated about the same when no reasons were provided. Results of the models are summarized in Table 5.2.

Our data do not show evidence that a reduction of the listener's inferred prior can result in a net reduction of his posterior belief, too. However, the data are still amenable to a pragmatic analysis. While the **prompted** and **unprompted** items received similar ratings when no reasons were provided, only the **prompted** items saw a net increase in the posterior when reasons were provided. Since the reasons did not pragmatically reduce the prior for the **prompted** items, their evidential strength resulted in a net increase in the posterior. However, the reasons did pragmatically reduce the prior for **unprompted** items (as per RQ1), so their evidential strength was sufficient to recover from that reduction, but not sufficient to see any further improvement.

It should be noted that, while our RSA model does predict a reduction of the posterior when too many reasons are provided, this effect can be quite small, depending on the selection of hyperparameters (utterance cost, sample size, rationality, etc). Even if the effect is visible on a continuous probability density function (i.e. Figure 3.5), it need not be large enough to reach significance when discretized to a 7-point Likert scale (as evidenced by the simulation plotted in Figure 5.2, center panel). The critical evidence for the pragmatic model is that, although both **prompted** and **unprompted** items had similar ratings when no reasons were provided, the reasons increased the posterior *only in prompted contexts*. The evidential model predicts a similar increase across contexts for items with similar priors (as per Figure 5.2, right panel).

5.4 Notes on the free-text data

Although full quantitative analysis of the free-text responses is beyond the scope of this paper, it is worth noting that some participants commented that it was difficult to rate the favorability of the actions when no reasons were given. This suggests that some **null-0R** items may have been interpreted as *under*informative. Future work based on these comments will be proposed in Chapter 6.

At the end of the survey, participants were asked to report what they thought experiment was investigating. The vast majority of respondents did not mention explanation/justification at all. Many reported that they could not even guess what the experiment was about. Others suggested that it might be testing how the text-message medium affects tone recognition. These results indicate that participants did not trivially identify the research questions and attempt to give “correct” responses, providing validation for the experiment design.

Chapter 6

General Discussion

6.1 Summing up

In this study, we set out to explain the intuition that arguments sometimes seem weaker when more reasons are provided. This intuition seems contrary to basic principles of rational belief revision, which predict that argument strength should increase monotonically as additional favorable evidence becomes available. We proposed that this effect might be produced by social reasoning about the expected behavior of an efficient communicator. Adapting the Rational Speech Act framework to suit argumentative goals, we provided a pragmatic account of the "protest too much" effect, and tested the predictions of our model against a first-order Bayesian model.

Our experiment confirmed that listeners do indeed make inferences about speakers' motivations for providing reasons, and that those inferences do diminish the evidential effect of the reasons on listeners' posterior beliefs, compared to cases where the inference is absent. However, the inference we observed from excess reasons was not strong enough to result in a credible net *reduction* of the posterior Likert ratings, which the more fine-grained RSA model had predicted.

6.2 Limitations and future work

Our model employs a number of simplifications, which may limit its generalizability, and which make a robust "non-monotonicity" effect more difficult to observe experimentally. Firstly, we restrict the number of reasons a speaker may provide to 0 or 4. While 4 reasons does seem to be over-informative in the scenarios we tested (as evidenced by the observed inference to a reduced prior μ), 0 reasons may well be under-informative, which seems to be corroborated by the free-text data. In certain cases, an utterance which cites (perhaps) 1 reason might be nearly as evidentially strong as the 4-reason utterance, and also nearly as cost-effective as the 0-reason utterance. A follow-on study testing 1 vs 4 reasons, then, might show a more visible decrease in ratings between an appropriately informative utterance and an over-informative utterance.

Furthermore, we limited the set of possible priors that the listener might have to beta distributions with a fixed sample size parameter. Even restricting the priors to beta distributions is surely a vast oversimplification of agents' belief states, but in particular, fixing the sample size at a constant value obscures the fact that reasons can be more or less informative for different listeners with the same prior mean, if those listeners hold their opinion with different degrees of confidence. In a more realistic model, the pragmatic agents should infer both beta parameters, because the quantity of information gained by conditioning on a reason crucially depends on the "curiosity" or "stubbornness" of the listener's prior, in addition to its overall expected value.

Finally, we selected our positive linear likelihood function because of its formal behavior: it enforces our constraint that the reasons should count in favor of the conclusion. However, we used the same likelihood function to represent the evidential contribution of all of the reasons, across all items. While this simplification should not meaningfully affect the qualitative predictions of the model, a norming study that measures the evidential strength of each specific reason is required for a more accurate quantitative fit of our model to the data.

6.3 Concluding remarks: cooperativity and vigilance

The cooperative principle has long been essential to pragmatic theory, but in recent years, there has been a surge of interest in expanding Grice's insights to mixed-motive and non-cooperative communication (Cummins, 2025). Much of this literature takes inspiration from Sperber et al.'s (2010) proposal that we have "a suite of cognitive mechanisms for epistemic vigilance", by which we evaluate the credibility of our interlocutors' utterances. Significant progress has been made in experimentally verifying and computationally modeling the ways in which a presumption of vigilance affects what speakers say, and what listeners infer (Franke et al., 2020; Oey et al., 2023; Oktar et al., 2024; Zheng et al., 2026). Our study contributes a new piece to this puzzle.

Sperber et al. situate their theory within the debate on the epistemic status of testimony. The vigilance mechanisms that protect us from being deceived, they claim, enable us to rationally access vast amounts of knowledge which cannot be acquired just by perception and inference. Our study shows a surprising significance of Grice's Quantity maxim in this discussion. Even when we are able to make information available more "convincingly" to our interlocutors on the basis of inference from evidence, it seems that we may prefer to deploy this strategy pragmatically. When we optimize our utterances for the coordination of knowledge under cognitive constraints, it might sometimes be more "rational" to depend on trust after all.

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